SAT Unit 3.4
Models and Scatterplots

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Overview

Linear vs Exponential Model

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Linear Models

Definition: A linear model shows a **constant rate of change** — the dependent variable increases or decreases by the same amount for every unit increase in the independent variable.

$$y = mx + b$$

Where:

- m is the slope (rate of change)
- b is the y-intercept (starting value)

Key Characteristics:

- Graph is a straight line.
- Change is additive.
- Example: Earning \$15/hour $\rightarrow$ $y = 15x$

Exponential Models

Definition: An exponential model shows a **constant percentage rate of change** — the quantity increases or decreases by the same **proportion** over equal intervals.

$$y = a \cdot b^x$$

Where:

- a is the initial value
- b is the growth $(b > 1)$ or decay $(0 < b < 1)$ factor

Key Characteristics:

- Graph is curved — increasing/decreasing rapidly.
- Change is multiplicative.
- Example: Population doubles each hour $\rightarrow y = 100 \cdot 2^x$

Exponential Decay – Interpretation of Parameters

Question:
The function $M(t) = 4000(0.6)^{\frac{t}{3}}$ models the mass, in nanograms, of a particle after t years.
Which of the following is the best interpretation of $(0.6)^{\frac{t}{3}}$ in this context?

- **A.** The mass of the particle decreases by 60% every 4 months.
- **B.** The mass of the particle decreases by 60% every 3 years.
- **C.** The mass of the particle decreases by 40% every 4 months.
- **D.** The mass of the particle decreases by 40% every 3 years.

Answer: Decay Rate over Time

Explanation:
The exponential function is of the form:

$$M(t) = 4000 \cdot (0.6)^{\frac{t}{3}},$$

which means that every 3 years, the mass is multiplied by 0.6 — a 40% decrease from the previous amount.
Correct Answer: $\boxed{D}$

Comparing Linear vs Exponential Models

Linear: Constant difference **Exponential:** Constant ratio

Feature	**Linear**	**Exponential**
General Form	$y = mx + b$	$y = a \cdot b^x$
Rate of Change	Constant (add/subtract)	Constant (multiply/divide)
Graph Shape	Straight line	Curve (growth/decay)
Example	\$10 per day	Double each day
Real-World Use	Salaries, fees	Populations, investments

Key Insight: If the rate of change is constant **difference**, it's linear. If the rate of change is constant **ratio**, it's exponential.

Question: Linear Functions – Percentage Relationship

For $x > 0$, the function f is defined as follows: $f(x)$ equals 201% of x. Which of the following could describe this function?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear

Answer Explanation: Linear Functions – Percentage Relationship

Since 201% of x means $f(x) = 2.01x$, this is a linear function with a positive slope, so it increases at a constant rate.

Correct Answer: $\boxed{D}$

Practice: Identify the Model Type

A population of bacteria is observed to grow as follows:

Time (hours)	Population
0	100
1	200
2	400
3	800

Question: Which type of model best describes this data?

A. Linear Model

B. Exponential Model

C. Neither

Answer Explanation: Identify the Model Type

The population doubles each hour: $100 \rightarrow 200 \rightarrow 400 \rightarrow 800$. This shows a **constant ratio** (multiply by 2), so the model is $P(t) = 100 \cdot 2^t$.
Correct Answer: $\boxed{B}$

Scatterplots and Regression

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Understanding Scatterplots

Definition: A **scatterplot** is a graph that displays pairs of data points (x, y) as dots in the coordinate plane.

Purpose:

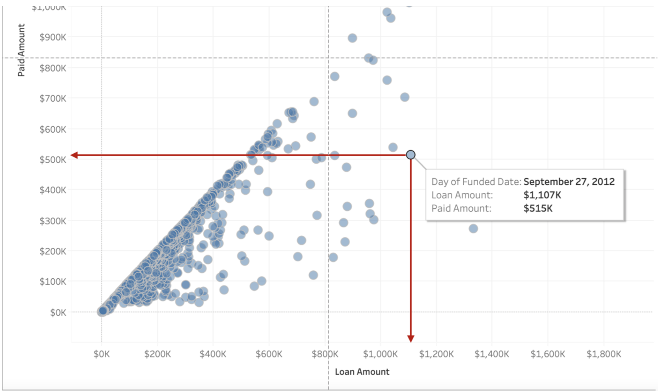
- Visualizes the relationship between two numerical variables.
- Reveals trends, clusters, outliers, or correlation direction.

Types of Trends:

- **Positive association:** As x increases, y increases.
- **Negative association:** As x increases, y decreases.
- **No association:** No visible pattern or relationship.

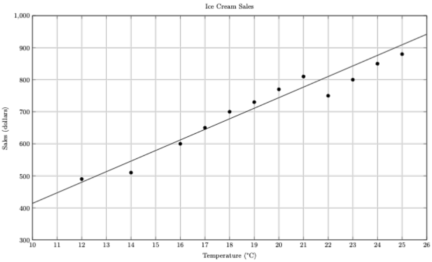
Note: Scatterplots do not imply causation, only correlation.

Scatterplots and Linear Regression



Linear regression is a line of best fit through a scatterplot: $y = mx + b$, where m is slope and b is the y-intercept. It minimizes squared vertical distances (residuals) and is used for prediction. The closer the points are to the line, the stronger the linear relationship.

Question: Line of Best Fit from Scatterplot



The scatterplot above shows a company's ice cream sales d, in dollars, and the high temperature t, in degrees Celsius (°C), on 12 different days. A line of best fit for the data is also shown.

Which of the following could be an equation of the line of best fit?

A. $d = 0.03t + 402$

B. $d = 10t + 402$

C. $d = 33t + 300$

D. $d = 33t + 84$

Answer Explanation: Line of Best Fit from Scatterplot

From the scatterplot:

- The line of best fit passes through approximately $(12, 480)$ and $(24, 876)$
- Estimating the slope:

$$\text{slope} = \frac{876 - 480}{24 - 12} = \frac{396}{12} = 33$$

- Plug into point-slope form:

$$d - 480 = 33(t - 12) \Rightarrow d = 33t + 84$$

Correct Answer: $\boxed{D}$

Practices

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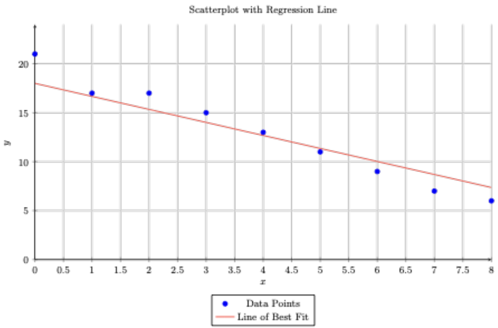
Question: Slope of Line of Best Fit



Which of the following is closest to the slope of the line of best fit shown?

A. -2.16

B. -1.38

C. -0.72

D. -0.46

Answer Explanation: Which of the following is closest to the slope of the...

Step 1. Find two points on the line of best fit:

- At $x = 1$, $y \approx 17$.
- At $x = 4$, $y \approx 13$.

Step 2. Compute the slope:

$$\text{slope} = \frac{13 - 17}{4 - 1} = \frac{-4}{3} \approx -1.33.$$

Step 3. Choose the closest answer:

$$-1.33 \text{ is closest to } -1.38.$$

Correct Answer: $\boxed{B}$

Question: An online service currently has 8,400 users. A random. . .

An online service currently has 8,400 users. A random sample of 350 users was selected and asked whether they will renew their subscription to the service. Of those surveyed, 12% stated that they will not renew their subscription. Based on this survey, which of the following is the best estimate of the total number of users who will not renew their subscription?

A. 42

B. 200

C. 1,008

D. 7,392

Answer Explanation: An online service currently has 8,400 users. A random...

Step 1. Calculate the estimated percentage of non-renewing users:

$$0.12 \times 8,400 = 1,008.$$

Correct Answer: $\boxed{C}$

Question: The employees at a new bookstore must stock a certain...

The employees at a new bookstore must stock a certain number of shelves so that the store is ready for its opening in two weeks. The employees stock shelves at a constant rate throughout the two weeks. If $p(t)$ is the number of shelves left to be stocked after t days, which of the following statements best describes the function p?

A. The function p is an increasing exponential function.

B. The function p is a decreasing exponential function.

C. The function p is an increasing linear function.

D. The function p is a decreasing linear function.

Answer Explanation: The employees at a new bookstore must stock a certain...

Step 1. Since shelves are stocked at a constant rate, the rate of decrease in the number of shelves left to stock is constant. That means $p(t)$ is decreasing linearly.

Correct Answer: $\boxed{D}$

Question: For the function f, the value of $f(x)$ is equal. . .

For the function f, the value of $f(x)$ is equal to half the value of x. Which of the following is the best description of this function?

A. Decreasing linear

B. Decreasing exponential

C. Increasing linear

D. Increasing exponential

Answer Explanation: For the function f, the value of $f(x)$ is equal...

Step 1. The function is $f(x) = \frac{1}{2}x$.
Step 2. Since the function is a constant multiple of x, it is a linear function.
Step 3. The coefficient $\frac{1}{2}$ is positive, so the function is increasing.
Correct Answer: $\boxed{C}$

Thank You!

We appreciate your time and focus.

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